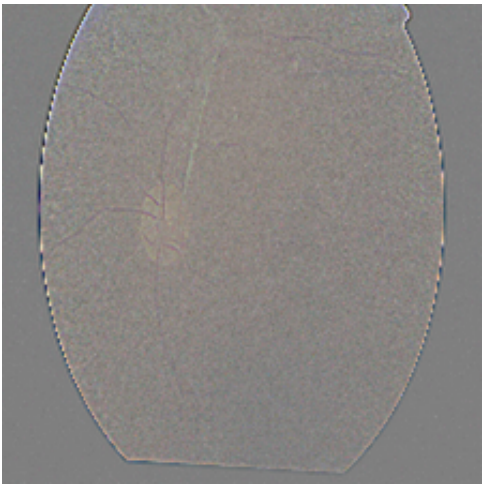


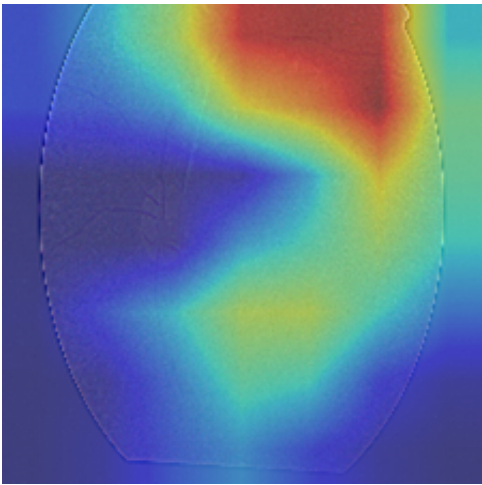
Clinical Diagnostic & RAG Report — Diabetes

Target Domain:	Diabetes	Image File:	PIL_Image
Predicted Class:	Proliferate DR	Confidence:	49.4%
95% Conformal Set:	['Mild DR', 'Moderate DR', 'Proliferate DR']	Uncertainty Status:	■■ Human Review Recommended

Visual Diagnostic Imaging & Grad-CAM Heatmap



Original Medical Scan



Grad-CAM Saliency Overlay (ROI)

AI Clinical Explanation (FLAN-T5 Grounded RAG)

"The image was classified as 'Proliferate DR' with 49.4% confidence. Biomedical literature (PMID:34488028) indicates: Diabetic retinopathy (DR) has become one of the major causes of blindness. Due to the increased prevalence of diabetes worldwide, diabetic patients exhibit high probabilities of developing DR. There is a need to develop a labor-less computer-aided diagnosis system to support the clinical diagnosis."

Faithfulness & Quantitative Text Metrics

NLI Faithfulness Score:	0.2660 / 1.0000	ROUGE-1:	0.7917
BLEU-4:	0.7455	BERTScore Similarity:	0.5284

Retrieved PubMed Literature Evidence

[1] Simple methods for the lesion detection and severity grading of diabetic retinopathy by image processing and transfer learning. (PMID: 34488028)

"Diabetic retinopathy (DR) has become one of the major causes of blindness. Due to the increased prevalence of diabetes worldwide, diabetic patients exhibit high probabilities of developing DR. There is a need to develop a labor-less computer-aided diagnosis system to support the clinical diagnosis."

[2] Automatic severity grade classification of diabetic retinopathy using deformable ladder Bi attention U-net and deep adaptive CNN. (PMID: 37338737)

"Long-term exposure to diabetes mellitus leads to the formation of diabetic retinopathy (DR), which can cause vision loss in working-age adults. Early stage diagnosis of DR is highly essential for preventing vision loss and preserving vision in people with diabetes. The motivation behind the severity grade classification of DR is to develop an automated system that can assist ophthalmologists and healthcare professionals in the diagnosis and management of DR."

[3] Advanced CNN Deep Learning Model for Diabetic Retinopathy Classification. (PMID: 42292681)

"BACKGROUND: Diabetic Retinopathy (DR) is one of several retinal microvascular complications of Diabetes Mellitus (DM), a disease of increasing global prevalence. However, early detection and treatment can reduce or even prevent DR progression. In this work, Deep Learning (DL) techniques are used to grade DR from an early stage using either binary or multiclass classification as a clinical aid to help reduce the risk of patient vision loss."

[4] Automated Detection and Diagnosis of Diabetic Retinopathy: A Comprehensive Survey. (PMID: 34460801)

"Diabetic Retinopathy (DR) is a leading cause of vision loss in the world. In the past few years, artificial intelligence (AI) based approaches have been used to detect and grade DR. Early detection enables appropriate treatment and thus prevents vision loss."

[5] Automated Detection and Diagnosis of Diabetic Retinopathy: A Comprehensive Survey. (PMID: 34460801)

"In addition, a comprehensive list of available DR datasets is reported. This list was constructed using both the PICO (P-Patient, I-Intervention, C-Control, O-Outcome) and Preferred Reporting Items for Systematic Review and Meta-analysis (PRISMA) 2009 search strategies. We summarize a total of 114 published articles which conformed to the scope of the review."